

USER'S GUIDE FOR

Global Rainfall Map in Near-Real-Time by JAXA Global Rainfall Watch

(GSMaP_NRT)

Version3.0

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Earth Observation Research Center
Japan Aerospace Exploration Agency

1. Introduction

We offer hourly Global Rainfall Map in Near-Real-Time (GSMaP_NRT), available four hours after observation, using the JAXA Global Rainfall Watch System. The system is based on the combined MW-IR algorithm using GPM-Core GMI, TRMM TMI, GCOM-W1 AMSR2, DMSP series SSMIS, NOAA series AMSU, MetOp series AMSU, and Geostationary IR, developed by the GSMaP (Global Satellite Mapping of Precipitation) project.

The newly developed algorithm for the Global Precipitation Measurement (GPM) mission (GPM-GSMaP Ver.6) is used to retrieve rain rate, and rainfall product is the same to the GPM Global Rainfall Map product distributed from the JAXA G-Portal (<https://www.gportal.jaxa.jp>). GPM-GSMaP Ver.6 is the latest algorithm by the Global Satellite Mapping of Precipitation (GSMaP) project, and it is based on the heritage of the study "Production of a high-precision, high-resolution global precipitation map using satellite data," sponsored by Core Research for Evolutional Science and Technology (CREST) of the Japan Science and Technology Agency (JST) during 2002-2007. Since 2007, GSMaP project activities are promoted by the JAXA Precipitation Measuring Mission (PMM) Science Team).

The main feature of the GSMaP algorithm is utilization of various attributes derived from TRMM Precipitation Radar (PR) that is the first spaceborne precipitation radar. Addition of the GPM Core Observatory Dual-frequency Precipitation Radar (DPR) to the database will be scheduled in next algorithm version up.

2. Description of rainfall data

Table 1 summarizes major specification of rainfall data.

Table 1 Description of rainfall data

Variable	Rainfall rate (mm/h), Satellite Information Flag, Observation Time Flag
Domain	Global (60N - 60S)
Grid resolution	0.1 degree latitude/longitude
Temporal resolution	1 hour
Data delay	4 hour after observation completed

3. Algorithms

For details of the GPM-GSMaP algorithm Ver.6, please see “Global Satellite Mapping of Precipitation (GSMaP) for GPM: Algorithm Theoretical Basis Document (ATBD)” from JAXA/EORC GPM Web Site (http://www.eorc.jaxa.jp/GPM/doc/algorithm_e.htm).

1) Microwave Imager Algorithm

GSMaP microwave imager (MWI) algorithm retrieves global precipitation rates from satellite microwave imager (TMI, GMI, AMSR, AMSRE, AMSR2, SSM/I) bright temperatures (TBs). To develop this algorithm, we improved that of Aonashi et al. (2009) by introducing the orographic rain adjustment method etc..

The basic idea of the MWI algorithm is to find precipitation rates that give radiative transfer mode (RTM)-calculated TBs that best fit with TMI TBs. The MWI algorithm employs polarization corrected temperatures (PCTs) at higher frequencies (37 and 85 GHz for TMI) over land and coast, TBs with vertical polarization at lower frequencies (10, 19, and 37 GHz for TMI) in addition to the higher frequency PCTs over ocean. (For SSM/I, this algorithm employs TB polarization differences at 19 GHz instead of TBs at 10GHz.)

The MWI algorithm consists of the forward calculation part to calculate the LUTs, and the retrieval part to estimate precipitation rates from the observed TBs using the LUTs.

In the forward calculation part, we calculate LUTs for homogeneous precipitation by incorporating atmospheric and surface variables of the Japan Meteorological Agency (JMA) global analyses or forecasts and precipitating cloud models based on TRMM observation etc. (Takayabu, 2008; Takahashi and Awaka, 2005) into the RTM program of Liu (1998). We also calculate LUTs for orographic precipitation, following Shige *et al.* (2013) and Taniguchi *et al.* (2013). We then derive LUTs for inhomogeneous precipitation from the above LUTs using the approximation of Aonashi and Liu (2000).

The retrieval part performs a detection of precipitation, a retrieval using scattering signals, and an over-ocean retrieval using emission signals. For the detection of precipitation, we adopt the methods of Seto *et al.* (2005), Kubota *et al.* (2007), and Kida *et al.* (2009) over land, coastal areas, and ocean, respectively. In the retrieval using scattering signals, we employ dual frequency PCTs (at 37 and 85 GHz for TMI). We introduce an adjustment method using indices of frozen precipitation depth and surface temperature. In addition, we detect orographic precipitation areas where we use LUTs for orographic precipitation (Yamamoto and Shige, 2014). In the over-ocean retrieval using emission signals, we derive precipitation rate by minimizing a cost function for lower frequency, vertically-polarized TBs (10, 19, and 37 GHz for TMI) with the scattering retrievals as the first guess.

2) Microwave Sounder Algorithm

The GSMaP over-ocean rainfall retrieval algorithm for microwave sounders (MWS) was developed by Shige *et al.* (2009) based on the GSMaP algorithm for microwave imagers (MWI). This algorithm combines an emission-based estimate from brightness temperature (Tb) at 23 GHz and a scattering-based estimate from Tb at 89 GHz. Precipitation inhomogeneities are also taken into

account. Improvements of rain/no-rain classification method over coast and mitigation of underestimation in mountainous areas (Shige et al., 2013, Taniguchi et al., 2013) have been done.

The GSMaP over-land rainfall retrieval algorithm for MWS is a scattering algorithm that mainly relies on Tb data at 89 GHz and its rain/no-rain classification method has been developed utilizing Tb data at 150 GHz and 180 GHz (Kida et al. 2012).

In the MWS algorithm, LUT is fitted by a five degree polynomial function of incident angle number. The MWS algorithm saves coefficients of the regression function. The LUT is reconstructed from the regression coefficients in the retrieval algorithm for MWS.

3) Microwave Imager/Sounder Algorithm

The GSMaP rainfall retrieval algorithm for microwave imager/sounders (MWIS) was developed by Kubota *et al.* (2011) based on the GSMaP algorithm for microwave imagers (MWI). SSMIS does not have 10 GHz channels same as SSM/I, the algorithm combines emission-based estimates using TB polarization differences at 19 GHz, in addition to emission-based estimates using TB at 19 GHz vertical polarization and scattering-based estimates using TB at 89 GHz, over the ocean. Precipitation inhomogeneities are also taken into account. See Kubota *et al.* (2011) for more details.

The GSMaP over-land rainfall retrieval algorithm for MWIS was developed based on the GSMaP algorithm for microwave imagers (MWI).

4) Microwave-IR Combined Algorithm

GSMaP_MVK is a product that integrates passive microwave radiometer data with infrared radiometer data in order to have high temporal (1 hour) and spatial (0.1 degree) resolution global precipitation estimates. The product (GSMaP_MVK) is produced based on a Kalman filter model that refines the precipitation rate propagated based on the atmospheric moving vector derived from two successive IR images. The detail of the algorithm can be found in Ushio *et al.* (2009).

4. Algorithm inputs

1) Geostationary satellite data

Until 22Z 28 March 2012, pixel-resolution data of MTSAT (operated by Japan Meteorological Agency), METSOSAT-7/-8 (operated by EUMETSAT), GOES-11/-12 (operated by NOAA), provided by the Japan Weather Association (JWA) was used. [Satellite zenith angle correction method](#), which is developed and distributed by NOAA Climate Prediction Center (CPC), has been applied in merging each IR data.

Since 23Z 28 March 2012, [Globally-merged, full-resolution \(~4km\) IR Data](#), which is merged from the ~11 micron IR channels aboard the MTSAT, METSOSAT-7/-8, and GOES-11/-12, produced by NOAA/CPC, has been used.

The range of the latitude is 60N-60S. The temporal resolution is 1 hour in this product.

2) Low Earth Orbit satellite data

Table 2 Summary of input low Earth orbit data

Satellite	Height (km)	Instrument	Category	frequency (GHz)	Note	
GPM Core	407	GMI	imager	10.7, 19.4, 21.3, 37, 85.5, 166, 183.31±3, 183.31±7	Introduced into GSMaP_NRT since 3 Sep. 2014	
TRMM	402	TMI	imager	10, 19, 21, 37, 85		
AQUA	705	AMSR-E	imager	7, 10, 19, 24, 37, 89	Not operational since 4 Oct. 2011	
GCOM-W1	705	AMSR2	imager	7, 10, 19, 24, 37, 89	Introduced into GSMaP_NRT system since 1 Jul. 2013	
DMSP-F13	833	SSM/I	imager	19, 22, 37, 85	Not operational since 18 Nov. 2009	
DMSP-F14	833	SSM/I			Not operational since 24 Aug. 2008	
DMSP-F15	833	SSM/I			Only rain over the ocean has been used since Aug. 2006. Not used since 3 Sep. 2014	
DMSP-F16	833	SSMIS	imager/sounder	19.4, 22.2, 37, 91.7, 60-63, 50-59, 150, 183.31±1, 183.31±3, 183.31±7	Introduced into GSMaP_NRT system since 11 Jun. 2010	
DMSP-F17	850	SSMIS			Introduced into GSMaP_NRT system since 1 Jul. 2013	
DMSP-F18	850	SSMIS				
NOAA-N18	870	AMSU-A/MHS	sounder	23.8-89.1 (AMSU-A), 89, 157, 183.311±3, 183.311±5, 190.311 (MHS)	Introduced into GSMaP_NRT since 3 Sep. 2014	
NOAA-N19					Introduced into GSMaP_NRT system since 1 Aug. 2011	
MetOp-A	817					Introduced into GSMaP_NRT since 3 Sep. 2014
MetOp-B						

3) Ancillary Data

- JMA Global Forecast and Global Analysis (GANAL), 6-hourly, 0.5 degree grid box since 2011
- JMA Merged satellite and in situ data Global Daily Sea Surface Temperatures (MGDSST), daily, 0.25 degree grid box

5. Difference from previous GSMP rainfall products (V5.222)

Following improvements in the algorithm was implemented in Ver.6.

- Calculation of LUTs in 6-hourly
- Use of JMA Global Forecast data for calculation LUT of GSMP_NRT
- Improvements in microwave imager algorithm based on AMSR2 precipitation standard algorithm, including new land algorithm, new coast detection scheme, etc.
- Development of orographic rainfall correction method for warm rainfall in coastal area
- Update of database, including rainfall detection over land, land surface emission database developed by Japanese DPR/GMI combined team, etc.
- Development of microwave sounder algorithm over land

6. Some caveats for data users

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- Anyone wishing to publish any results using the data from the JAXA Global Rainfall Watch should clearly acknowledge the ownership of the data in the publication (for example, 'Global Rainfall Map in Near-Real-Time (GSMP_NRT) by JAXA Global Rainfall Watch' was produced and distributed by the Earth Observation Research Center, Japan Aerospace Exploration Agency). If you have benefited from GSMP rainfall products, please cite the major papers listed in Section 8 that you have benefited from.
- We would appreciate receiving a preprint and/or reprint of publications in which you utilize our data. Relevant publications should be sent to:
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8. Papers describing the GSMP project

Papers describing the GSMaP project and algorithm are as follows.

(About GSMaP project)

- K. Okamoto, T. Iguchi, N. Takahashi, K. Iwanami and T. Ushio, 2005: The global satellite mapping of precipitation (GSMaP) project. *25th IGARSS Proceedings*, 3414-3416.
- K. Okamoto, T. Iguchi, N. Takahashi, T. Ushio, J. Awaka, S. Shige, and T. Kubota, 2007: High precision and high resolution global precipitation map from satellite data. *ISAP 2007 Proceedings*, 506-509.
- T. Kubota, S. Shige, H. Hashizume, K. Aonashi, N. Takahashi, S. Seto, M. Hirose, Y. N. Takayabu, K. Nakagawa, K. Iwanami, T. Ushio, M. Kachi, and K. Okamoto, 2007: Global Precipitation Map using Satelliteborne Microwave Radiometers by the GSMaP Project : Production and Validation. *IEEE Trans. Geosci. Remote Sens.*, **45(7)**, 2259-2275.

(About microwave imager algorithm)

- K. Aonashi, J. Awaka, M. Hirose, T. Kozu, T. Kubota, G. Liu, S. Shige, S., Kida, S. Seto, N. Takahashi, and Y. N. Takayabu, 2009: GSMaP passive, microwave precipitation retrieval algorithm: Algorithm description and validation. *J. Meteor. Soc. Japan*, **87A**, 119-136.
- A. Taniguchi, S. Shige, M. K. Yamamoto, T. Mega, S. Kida, T. Kubota, M. Kachi, T. Ushio, and K. Aonashi, 2013: Improvement of high-resolution satellite rainfall product for Typhoon Morakot (2009) over Taiwan. *J. Hydrometeor.*, **14**, 1859-1871.
- S. Shige, M.K. Yamamoto, and A. Taniguchi, 2014. Improvement of TMI rain retrieval over the Indian Subcontinent. *Geophys. Monogr. Ser.* (in print).
- M.K. Yamamoto, and S. Shige, 2014: Implementation of an orographic/nonorographic rainfall classification scheme in the GSMaP algorithm for microwave radiometers. *Atmos. Res.* (in print)

(About microwave sounder algorithm)

- S. Shige, T. Yamamoto, T. Tsukiyama, S. Kida, H. Ashiwake, T. Kubota, S. Seto, K. Aonashi and K. Okamoto, 2009: The GSMaP precipitation retrieval algorithm for microwave sounders. Part I: Over-ocean algorithm. *IEEE Trans. Geosci. Remote Sens*, **47**, 3084-3097.
- S. Kida, S. Shige, and T. Manabe, 2010: Comparison of rain fractions over tropical and sub-tropical ocean obtained from precipitation retrieval algorithms for microwave sounders. *J. Geophys. Res.*, **115**, D24101, doi:10.1029/2010JD014279.
- S. Kida, T. Kubota, M. Kachi, S. Shige, and R. Oki, 2012: Development of precipitation retrieval algorithm over land for a satellite-borne microwave sounder. *Proc. of IGARSS 2012*, 342-345.

(About microwave imager/sounder algorithm)

T. Kubota, S. Shige, M. Kachi, and K. Aonashi. 2011: Development of SSMIS rain retrieval algorithm in the GSMaP project. *Proc 28th ISTS*, 2011-n-46.

(About microwave-IR combined algorithm)

T. Ushio, T. Kubota, S. Shige, K. Okamoto, K. Aonashi, T. Inoue, N., Takahashi, T. Iguchi, M. Kachi, R. Oki, T. Morimoto, and Z. Kawasaki, 2009: A Kalman filter approach to the Global Satellite Mapping of Precipitation (GSMaP) from combined passive microwave and infrared radiometric data. *J. Meteor. Soc. Japan*, **87A**, 137-151.

T. Ushio, T. Tashima, T. Kubota, and M. Kachi, 2013: Gauge Adjusted Global Satellite Mapping of Precipitation (GSMaP_Gauge), *Proc. 29th ISTS*, 2013-n-48.

(About NRT system)

M. Kachi, T. Kubota, T. Ushio, S. Shige, S. Kida, K. Aonashi, and K. Okamoto, 2011: Development and utilization of “JAXA Global Rainfall Watch” system. *IEEJ Transactions on Fundamentals and Materials*, **131**, 729-737. (In Japanese with English abstract)

T. Ushio, and M. Kachi, 2009: Kalman filtering application for the Global Satellite Mapping of Precipitation (GSMaP). *Chapter for “Satellite Rainfall Applications for Surface Hydrology” (Edited by Mekonnen Gebremichael and Faisal Hossain)*, Springer, ISBN978-9048129140, 105-123.

Additional related papers are listed on the JST/CREST GSMaP Project Website
http://sharaku.eorc.jaxa.jp/GSMaP_crest/html/publications.html